

JSPM University Pune

Faculty of Science and Technology



NEP Aligned Syllabus for

B.Tech. First Year

(Effective from AY: 2024-25)



JSPM UNIVERSITY PUNE

Recognized by the UGC u/s 2 (f) of UGC Act 1956 and enacted by the
State Government of Maharashtra - JSPM University Act, 2022 (Mah. IV of 2023)

JSPM University Pune		
F.Y. B. Tech “Fundamentals of Python Programming”		
Course Type: SEC	Course Title: FPP	
Course Code: 240GCSB62_01	Teaching Scheme: 2 lectures/week	Examination Scheme:
Credits:3	Lecture (L): 2 Tutorial (T): Practical (P): Experiential Learning (EL):	CIE: ESE:
Prerequisite Courses, if any:		
Course Objectives: - To introduce students to the Python programming language and its features. - To develop students' understanding of basic programming constructs such as variables, data types, operators, and control flow. - To familiarize students with advanced concepts including functions, modules, classes, and object-oriented programming. - To enable students to work with file handling, regular expressions, and data serialization. - To teach students how to create graphical user interfaces (GUIs) using Python libraries. - To provide students with practical experience in data manipulation, numerical computing, and data visualization using Python libraries.		
Course Outcomes: Upon successful completion of the course, students will be able to: CO1: Write Python scripts to solve a variety of programming problems. CO2: Utilize fundamental programming constructs and data structures in Python. CO3: Design and implement object-oriented solutions using classes and inheritance. CO4: Handle file input/output operations and manage exceptions effectively. CO5: Work with modules and external libraries for extended functionality. CO6: Develop graphical user interfaces (GUIs) for interactive applications. CO7: Create web applications and scripts using Python for server-side processing.		



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Course Contents		
Unit I	Python Basics	(4 Hrs)
Introduction to programming languages, Python as a programming language, History of python, Python versions, Python installation, Executing python from the command line, Invoking python IDLE, Python documentation, Getting help, Dynamic types, Python reserved words, Naming conventions, Character set, Comments, Identifiers, Data types, Operators, Assigning values to variables, Type conversions, String methods, Simple output, Simple input: Input function, Mutable vs immutable objects in python, Lists: Create and access, Lists: Modify and slice, Lists: Operations, Lists: Methods, Sets: Create and operations, Sets: Operators and methods, Sets: Frozenset, Sets: Methods, Tuples: Create and access, Tuples: Slice and alteration, Dictionaries: Create and access, Dictionaries: Modify and delete, Dictionaries: Methods, Copying collections: Shallow, Copying collections: Deep.		
Mapping of Course Outcomes for Unit I	CO1: Write Python scripts to solve a variety of programming problems.	
Unit II	Language Concepts	(4 Hrs)
Indenting requirements, Control statements, Decision making statement: If statement, If statement, If-else statement, Implementation, If-elif-else statement, Implementation, Decision making statement: Nested if-else statement, Nested if-else statement, Implementation, Iteration statements: While loop, While loop, Implementation, Iteration statements: While loop with else, While loop with else, Implementation, Iteration statements: For loop, For loop, Implementation, For loop range function with else, Implementation, For loop with object sequences, Nested for loops, Implementation, Break statement, Implementation, Continue statement, Implementation, Functions, Functions: Built in functions, Functions: User-defined functions, Calling a function, Returning a value from the function, Implementation, Scope or lifetime of variables, Creation and usage of global variables, Creation and usage of nonlocal variables, Passing collections to a function, Variable number of arguments, Implementation, Keyword arguments, Optional parameters, Default parameters, Nested functions, Recursive functions, Advantage and disadvantage, Passing functions to a function, map() function, filter() function, Lambda functions		
Mapping of Course Outcomes for Unit II	CO2: Utilize fundamental programming constructs and data structures in Python.	



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Course Contents		
Unit III	OOP, Exceptions and I/O	(4 Hrs)
Object-oriented programming concepts, Class and object, Abstraction and encapsulation, Inheritance, Polymorphism, Classes in Python, Creating objects: Instance methods, Implementation, Memory management, Constructors, Constructors with parameters, Optional parameters in constructor, Deleting attributes and objects, Special methods, Class variables, Inheritance, Accessing base class element in derived class, Single inheritance, Multi-level inheritance, Multiple inheritance, Access modifiers, Polymorphism, Operator overloading, Method overloading, Method overriding, Python errors: Syntax errors, Built-in exceptions, Exception handling, Simple exceptions, Multiple exceptions, Using else and finally, Raise an exception, Assert statement, Data streaming and buffering: Serial data, I/O streams and buffers, Access modes, File open, File close, Exceptions in file, Writing to files, Reading from files, seek() and tell() methods, readline() and readlines() methods.		
Mapping of Course Outcomes for Unit III	CO3: Design and implement object-oriented solutions using classes and inheritance. CO4: Handle file input/output operations and manage exceptions effectively.	
Unit IV	Modules and Regular Expressions	(4 Hrs)
Modules in Python, Using modules in Python code, Import statement variances, Module search path, Loading and reloading of modules, dir() function, Python built-in modules: sys, Python built-in modules: math, Methods in module math, Python built-in modules: datetime, Python built-in modules: random, Regular expressions, Regular expression methods, Implementation, re.split(), Implementation, re.sub(), Implementation, re.subn(), re.search(), re.compile(), Implementation, Match object, Implementation, Raw string with "r" or "R" prefix. RegEx quantifiers, RegEx greedy match, RegEx non-greedy match, Greedy vs non-greedy match, RegEx flags.		
Mapping of Course Outcomes for Unit IV	CO5: Work with modules and external libraries for extended functionality.	



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Course Contents		
Unit V	Data Structures and GUI	(4 Hrs)
Abstract data structures, Primitive data structure, Non-primitive data structures, List comprehensions, Accessing elements, Performing operations, Comprehension using If, Comprehension If-else, Nested list comprehensions, Dictionary comprehensions, Accessing elements, Performing operations, Comprehension using zip(), Comprehension for lambda functions, Nested dictionary comprehensions or dictionaries with compound values, Processing lists in parallel, Time functionality: Big O notation, Case scenarios, Time complexity in python collections, GUI in Python, Components and events, GUI example, Widgets, Attributes for widgets, Label widget, Button widget, Entry widget. Combo box widget, Check button widget, Radio button widget, Canvas widget, Frame widget, Message box widget, Message widget, Menu widget, Methods, Menu button, Implementation, CGI basics, Configuring CGI, HTML form elements, Form structure, Python and CGI program, CGI scripts, HTTP headers and environment variables, GET method, POST method, Complete CGI code, Advantages and disadvantages.		
Mapping of Course Outcomes for Unit V	CO6: Develop graphical user interfaces (GUIs) for interactive applications. CO7: Create web applications and scripts using Python for server-side processing.	
Unit VI	Python Applications	(4 Hrs)
OS methods, Environment methods, Directory methods, File methods, Implementation, Implementation, Serialization and deserialization, The pickle module, Pickling and unpickling: Dictionary, Protocol formats, Picklable vs unpicklable types, Serializing attribute connections, Pickle exceptions, Networking fundamentals, Basic communication model, Network topologies, RING topology, STAR topology, MESH topology, TREE topology, Transmission mediums and modes, The client/server model, Client interaction with server, The socket module, The server-side, The client-side, Threaded server, Numpy: Overview, Numpy: Setup, Datatypes, Numpy: Basic operations, Binary operations, NumPy operations, Slicing and indexing, Matplotlib: Multi plots, 3D plots, Data processing with pandas, Invalid value, Processing strings, Indexing and selecting data, iloc(): Integer based, Column name access, Group-by operations, Iterating through groups, Aggregation, Transformations, Filtration.		
Mapping of Course Outcomes for Unit VI	CO5: Work with modules and external libraries for extended functionality.	



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JSPM University Pune		
F.Y. B.Tech. “Fundamentals of Python Programming Practical”		
Course Type: SEC	Lab Course Title: FPP-Practical	
Course Code: 240GCSB62_01	Teaching Scheme: 2 hrs./Week	Examination Scheme:
Credits: 3	Lecture (L): Tutorial (T): Practical (P): 2 Experiential Learning (EL):	Theory (TH): Practical (PR): 50 Oral (OR): 50
Tools / Languages: Python interpreter 3.9 and above. IDLE and any IDE like Jupyter, VS Code.		
List of Laboratory Experiments		
1.	a) Display your name, count length of it and display what will be your age in year. b) Write python Program to calculate arithmetic operation using type conversion	
2.	Write simple python program that demonstrates the use of lists, tuples, and dictionaries.	
3.	Write a function to calculate factorial of a number. Develop a program to compute binomial coefficient (Given N and R).	
4	Develop a Python program to check whether a given number is palindrome or not and also count the number of occurrences of each digit in the input number.	
5.	Develop a program to read the student details like name, USN and marks in 3 subjects. Display the student details, total marks and percentage with suitable messages	
6.	Write a Python program that demonstrates the use of a recursive function, nested functions, and functional programming techniques such as map, filter, lambda functions and print the result, which is the sum of the squares of the even number.	
7.	Write a python program by creating a class called Employee to store the details of Name, Employee_ID, Department and Salary, and implement a method to update salary of employees belonging to a given department.	

8.	Write a python program to accept a file name from the user and perform the following operations 1.Display the first N line of the file 2.Find the frequency of occurrence of the word accepted from the user in the file
9.	Write Python program covers basic functionalities provided by the math module.
10.	Demonstrates a common approach to validating email addresses using regex in Python.
11.	Develop a program to backing Up a given Folder (Folder in a current working directory) into a ZIP File by using relevant modules and suitable methods.
12.	Write Python code that creates a simple graphical user interface (GUI) application.
13.	Create a basic CGI script in Python that allows students to submit their name and grade through a form and displays a personalized message based on their input?
14.	Develop a program to create a DataFrame from a NumPy array with custom column names.
15.	Write a Python program to Demonstrate how to Draw a Bar Plot, Scatter Plot using Matplotlib.

Reference Books:

1. "Python Crash Course" by Eric Matthes
2. "Automate the Boring Stuff with Python" by Al Sweigart
3. "Learning Python" by Mark Lutz
4. "Python Programming: An Introduction to Computer Science" by John Zelle
5. "Python Cookbook" by David Beazley and Brian K. Jones
6. "Fluent Python: Clear, Concise, and Effective Programming" by Luciano Ramalho
7. "Effective Python: 90 Specific Ways to Write Better Python" by Brett Slatkin